

**Question:**

A 38-year-old man comes to the physician because of progressive pain and swelling of his left knee for the past 2 days. He has been taking ibuprofen for the past 2 days without improvement. Four days ago, he scraped his left knee while playing baseball. He has a 2-month history of progressive pain and stiffness in his back. The pain starts after waking up and lasts for 20 minutes. He has type 2 diabetes mellitus. His older sister has rheumatoid arthritis. He is 170 cm (5 ft 7 in) tall and weighs 91 kg (201 lb); BMI is 31.5 kg/m<sup>2</sup>. Temperature is 39°C (102.2°F), pulse is 90/min, and blood pressure is 135/85 mm Hg. Examination shows an erythematous, tender, and swollen left knee; range of motion is limited. There are abrasions over the lateral aspect of the left knee. The remainder of the examination shows no abnormalities. Laboratory studies show a leukocyte count of 13,500/mm<sup>3</sup> and an erythrocyte sedimentation rate of 70 mm/h. Which of the following is the most appropriate next step in management?

Answer Choices:

- A. Bone scan
- B. MRI of the left knee
- C. Blood cultures
- D. Arthrocentesis of the left knee

**Reasoning + Answer:**

<think>

The patient presents with progressive pain and swelling in the left knee , which started with a scrape injury . The patient has a family history of rheumatoid arthritis and symptoms of stiffness in his back . Considering the erythematous , tender , and swollen knee with limited range of motion and elevated ESR , we should consider rheumatologic causes like rheumatoid arthritis or septic arthritis .

</think>

<answer>B . MRI of the left knee </answer>

